

PREGNANCY TESTS

Headings

- ▼ PREGNANCY TESTS
 - ▼ 1. BIOLOGICAL TESTS
 - Disadvantages of Biological Tests
 - ▼ 2. IMMUNOLOGICAL TESTS
 - Requisites for the Test:
 - Test Procedure:
 - OBSERVATION AND RESULTS (AGGLUTINATION INHIBITION)
 - ADVANTAGES OF IMMUNOLOGICAL TESTS

- **Definition :** Diagnostic tests used to detect or confirm pregnancy.
- **Underlying Principle :** Based on determining the presence of **Human Chorionic Gonadotropin (hCG)** in the maternal urine.
- **Categories :**
 - a. **Biological Tests** (performed on experimental animals)
 - b. **Immunological Tests** (based on antigen-antibody reactions)

1. BIOLOGICAL TESTS

- Performed using **experimental animals**.
- Can be performed only **2 to 3 weeks after conception**.

Test Name	Animal Used	Procedure & Route	Observation & Time Frame
1. Aschheim-Zondek Test (<i>1st test invented</i>)	Immature female mice	2 mL of urine injected subcutaneously daily for 2 days.	• Mice killed after 5 days. • Ovaries examined for corpus lutea / hemorrhagic spots (indicates hCG-induced ovulation).
2. Kupperman Test	Immature female rat	• 2 mL urine injected subcutaneously OR	• Subcutaneous: Ovarian changes seen after 6 hours.

Test Name	Animal Used	Procedure & Route	Observation & Time Frame
<i>(Modification of AZ test)</i>		• Injected intraperitoneally.	• Intraperitoneal: Ovarian changes seen after 2 hours .
3. Friedman Test	Mature virgin female rabbit	10 – 15 mL of urine injected intravenously (into ear vein).	Ovaries examined for ruptured follicles / corpus lutea after 48 hours .
4. Hogben Test	Female South African toad (<i>Xenopus laevis</i>)	20 – 30 mL of urine injected into the dorsal lymph sac .	hCG stimulates extrusion of ova / ovulation within 12 hours .
5. Galli-Mainini Test	Male amphibian (toad/frog)	2 mL of urine injected into the subcutaneous/lymph space.	hCG causes expulsion of spermatozoa in urine/cloacal fluid within 2 hours .

Disadvantages of Biological Tests

- a. Require maintenance of animal houses and live experimental animals.
- b. Can only be performed after 2 to 3 weeks of conception.
- c. Results are delayed (require waiting between 2 to 48 hours).
- d. Involves sacrificing animals.

2. IMMUNOLOGICAL TESTS

- **Basis** : Based on **double antigen-antibody reactions** (agglutination-inhibition principle).
- **Commonest Test** : **Gravindex Test**.
- **Principle** : Determining the inhibition of agglutination of **latex particles** (or sheep RBCs) coated with hCG.

Requisites for the Test:

- i. **hCG Antiserum** : Obtained from rabbits immunized against human hCG (contains rabbit anti-hCG antibodies).
- ii. **Latex Particles / Sheep RBCs** : Coated with pure hCG molecules.
- iii. **Urine Sample** : Fresh morning urine sample from the woman suspected of being pregnant.

Test Procedure:

Step 1: 1 drop of hCG Antiserum + 1 drop of Urine → Mix well on glass slide



Step 2: Add 1 drop of hCG-coated Latex Particles

OBSERVATION AND RESULTS (AGGLUTINATION INHIBITION)

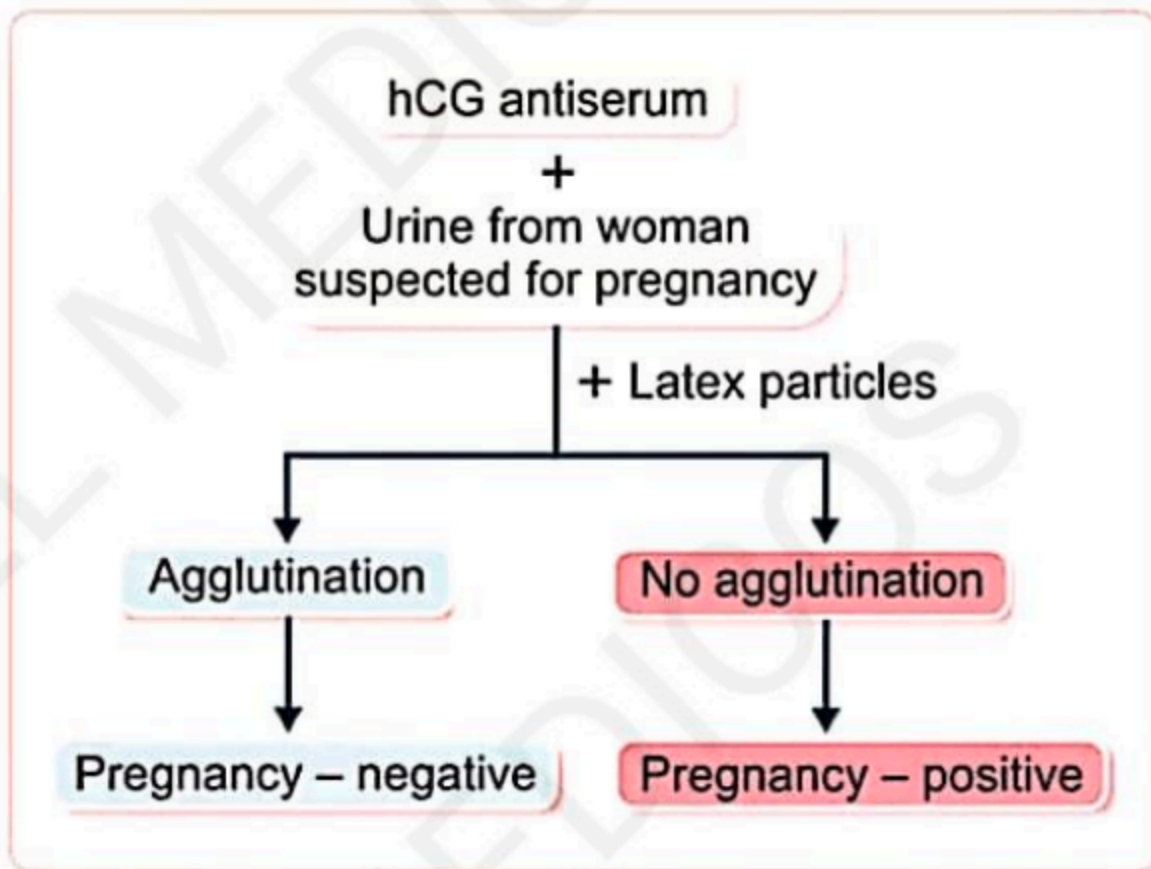
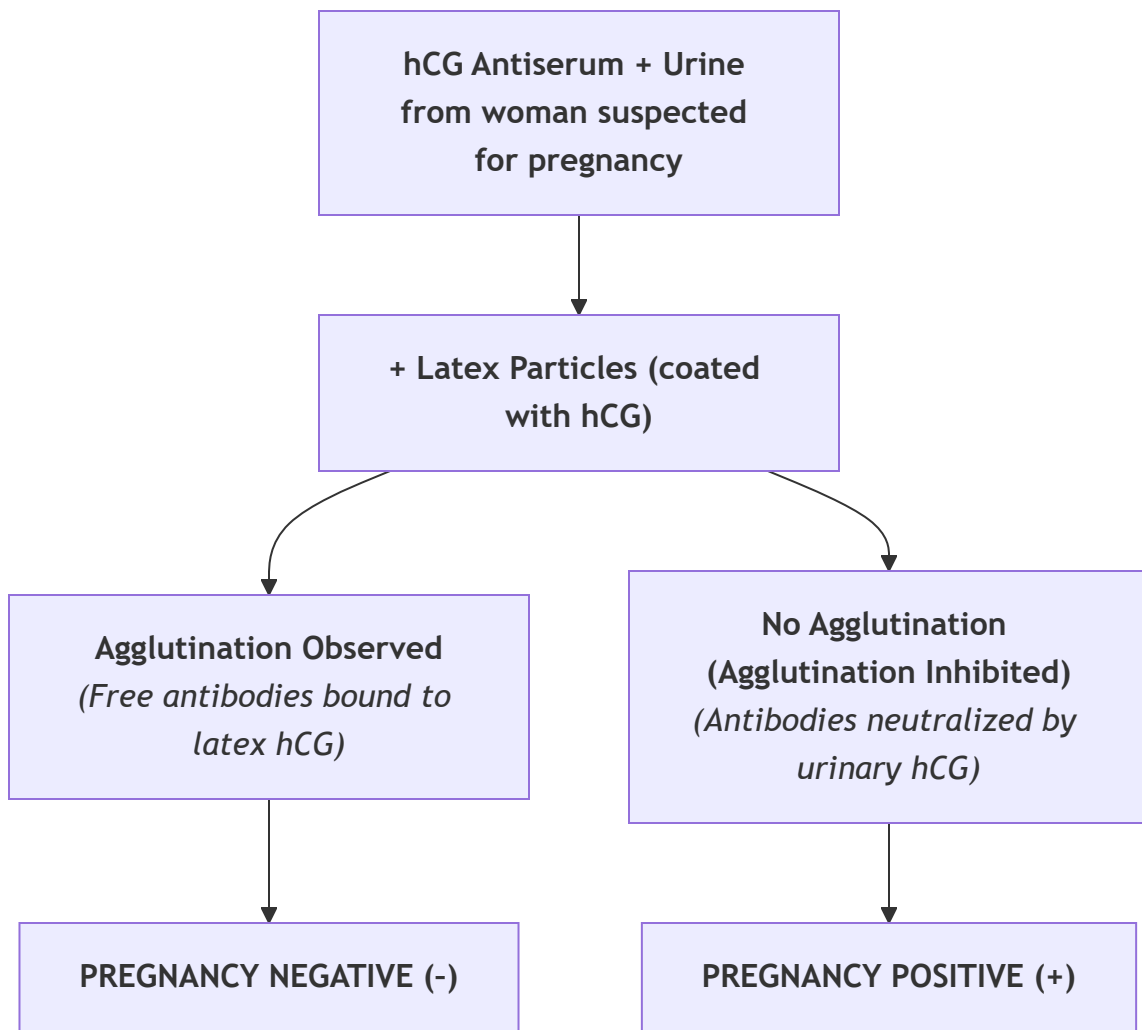


FIGURE 86.1: Immunological test for pregnancy



ADVANTAGES OF IMMUNOLOGICAL TESTS

- a. **High Accuracy** : Much more accurate, specific, and reproducible than biological tests.
- b. **Rapid Results** : Results are obtained within **a few minutes**.
- c. **Simplicity** : Easy to perform with minimal laboratory equipment.
- d. **Early Detection** : Can detect pregnancy as early as the **6th day of conception** (even within the first few days).
- e. **Convenience** : Highly sensitive, single-step methods available in the form of rapid **pregnancy test cards/cassettes**.